

Learn HTML: Beginner's Guide

Lesson 1: Introduction to HTML

Introduction to HTML – Study Notes

Key Concepts

1. **HTML = HyperText Markup Language** – the standard language for creating web pages.
2. **Markup vs. Programming** – HTML describes *what* a page contains; it doesn't contain logic or calculations.
3. **Document Structure** – every HTML file follows a basic skeleton: `<html>`, `<head>`, and `<body>`.
4. **Elements & Tags** – HTML is built from elements that are written with opening and closing tags (e.g., `<p> ... </p>`).
5. **Browser Rendering** – browsers read the HTML file, interpret the tags, and render the visual page for the user.

Summary

HTML (HyperText Markup Language) is the foundational markup language used to create the structure and content of web pages. Unlike programming languages, HTML does not perform calculations or make decisions; instead, it tells the browser how to organize text, images, links, and other resources on a page. Every HTML document starts with a **doctype declaration** (`<!DOCTYPE html>`) followed by the root `<html>` element, which encloses two main sections: the `<head>` and the `<body>`.

The `<head>` section contains meta information that is not displayed directly on the page, such as the page title, character encoding, links to CSS stylesheets, and scripts. The `<body>` section holds all the content that users see—headings, paragraphs, lists, images, links, and more. Understanding this basic structure is essential because it provides the scaffold on which all further HTML, CSS, and JavaScript are built.

Important Points

- **Doctype** (`<!DOCTYPE html>`) tells the browser to use HTML5 standards.
- The **root element** `<html>` wraps the entire document.
- `<head>` is for metadata:
 - `<title>` – text shown in the browser tab.
 - `<meta charset="UTF-8">` – defines character encoding.
 - `<link rel="stylesheet" href="styles.css">` – connects external CSS.
- `<body>` holds visible content: headings (`<h1>` `<h6>`), paragraphs (`<p>`), images (`<img>`), links (`<a>`), etc.
 - Tags are case insensitive in HTML5, but the convention is lower case.
 - Most elements have an opening tag `<tag>` and a closing tag `</tag>`; self closing tags (e.g., `<img />`) do not need a separate closing tag.

- Proper nesting is required: tags opened inside another must be closed before the outer tag (e.g., `<p><strong>text</strong></p>`).

Examples

Example 1 – Minimal HTML5 Document

```
``html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My First Page</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a simple HTML document.</p>
</body>
</html>
``
```

****Explanation****

- `<!DOCTYPE html>` enables HTML5 mode.
- `lang="en"` declares the language (helps accessibility & SEO).
- `<meta charset="UTF-8">` ensures characters display correctly.
- The `<title>` appears in the browser tab.
- Inside `<body>`, an `<h1>` heading and a `<p>` paragraph are displayed.

Example 2 – Adding an Image and a Link

```
``html
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Images & Links</title>
</head>
<body>
  <h2>My Favorite Photo</h2>
  
  <p>Want to learn more? Visit
    <a href="https://developer.mozilla.org/en-US/docs/Web/HTML" target="_blank">
      MDN Web Docs
    </a>.
  </p>
```

</body>

</html>

****Explanation****

- `<img>` uses `src` for the image file, `alt` for accessibility, and optional size attributes.
- `<a>` creates a hyperlink; `href` points to the URL, and `target="_blank"` opens it in a new tab.

Quick Review

1. What does the `<head>` element contain, and why is it important?
2. Which tag defines the visible main heading of a page?
3. Write the doctype declaration for an HTML5 document.
4. How do you link an external CSS file from the `<head>`?
5. Explain the purpose of the `alt` attribute on an `<img>` tag.

Further Reading

- ****HTML Elements Reference**** – full list of tags and their attributes (MDN).
- ****Semantic HTML**** – using meaningful tags like `<article>`, `<section>`, `<nav>`.
- ****HTML Forms**** – collecting user input with `<form>`, `<input>`, `<textarea>`.
- ****Accessibility Basics**** – ARIA roles, proper heading hierarchy, and alt text.
- ****Introduction to CSS**** – styling the structure you just created.

